Lec-12

Primitive type	Wrapper Class
boolean	Boolean
byte	Byte
char	Character
float	Float
int	Integer
long	Long
short	Short
double	Double

Wrapper Classes

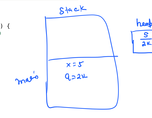
boolean
char
byte
short
int
long
float
double

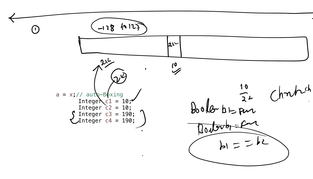
Autoboxing →

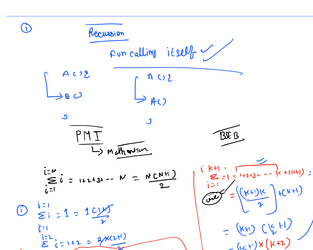
← Unboxing

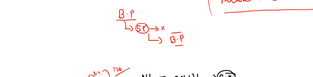
Boolean
Character
Byte
Short
Integer
Long
Float
Double

```
public class WrapperClass {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int x = 5;
        Integer a = 5;
        System.out.println(x);
        System.out.println(a);
    }
}
```



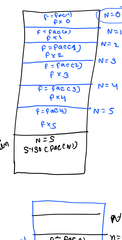






```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    System.out.println(Fac(n));
}

public static int Fac(int n) {
    int f = Fac(n - 1);// SP
    return f * n;// Self work
}
```



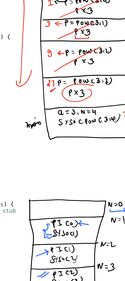

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    System.out.println(Fac(n));
}

public static int Fac(int n) {
    if (n == 0) {
        return 1;
    }
    int f = Fac(n - 1);// SP
    return f * n;// Self work
}
```

$a^n \rightarrow (q^{n-1}) \times a$

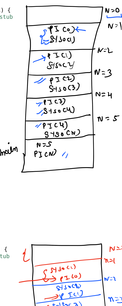
```
public class Power {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int a = 3;
        int n = 4;
        System.out.println(pow(a, n));
    }

    public static int pow(int a, int n) {
        if (n == 0) {
            return 1;
        }
        int p = pow(a, n - 1);
        return p * a;
    }
}
```

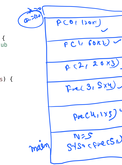



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    PI(n);
}

public static void PI(int n) {
    if (n == 0) {
        return;
    }
    PI(n - 1);
    System.out.println(n);
}
```




```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    PD(n);
}

public static void PD(int n) {
    if (n == 0) {
        return;
    }
    System.out.println(n);
    PD(n-1);
}
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    System.out.println(Fac(n, 1));
}

public static int Fac(int n, int ans) {
    if (n == 0) {

    }

    return Fac(n-1, ans*n);
}
```